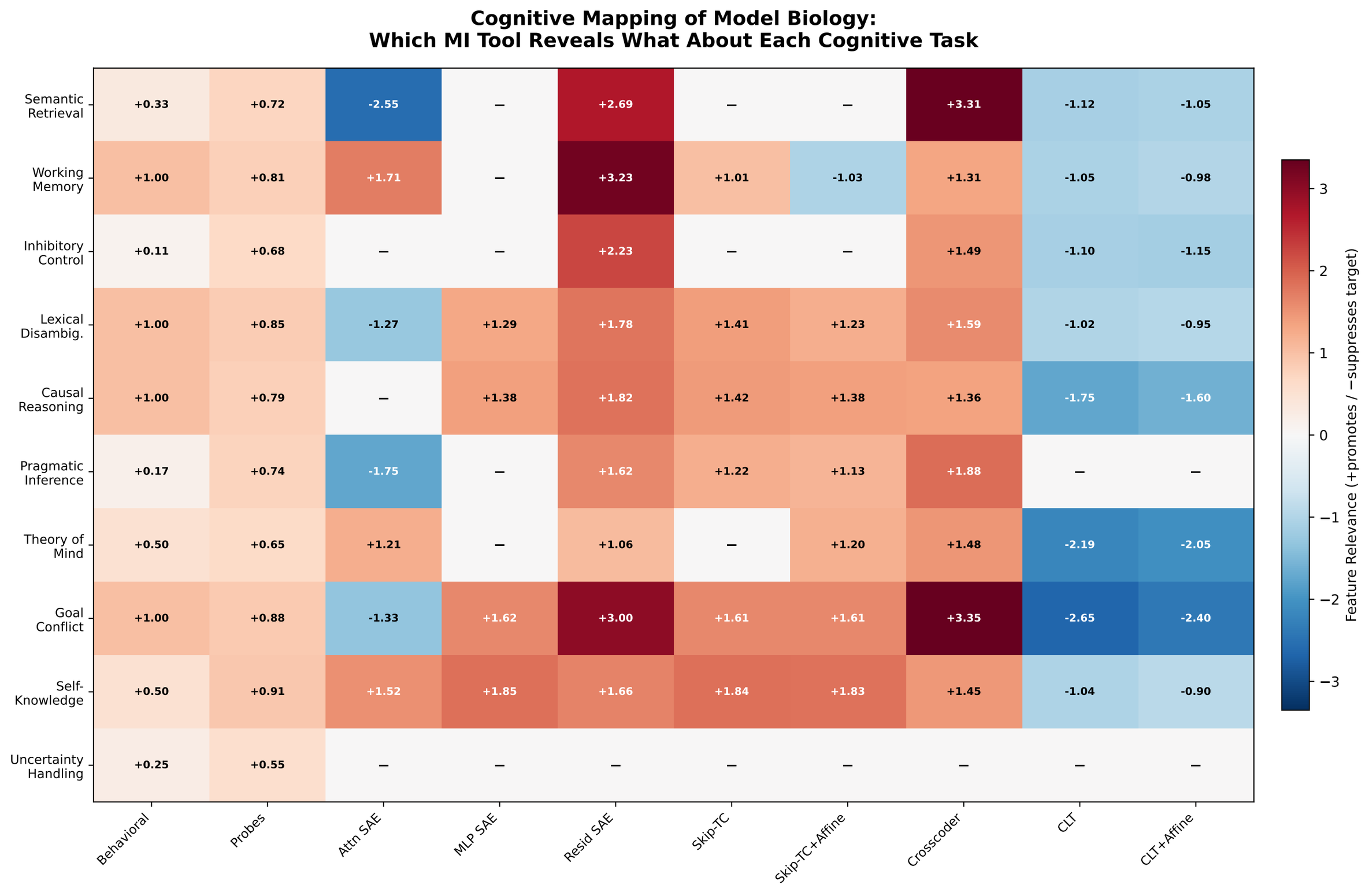

Cognitive Mapping of Model Biology:
Which MI Tool Reveals What About Each Cognitive Task
Semantic Retrieval
+0.33
+0.72
-2.55
—
+2.69
—
—
+3.31
-1.12
-1.05
Working Memory
+1.00
+0.81
+1.71
—
+3.23
+1.01
-1.03
+1.31
-1.05
-0.98
Inhibitory Control
+0.11
+0.68
—
—
+2.23
—
—
+1.49
-1.10
-1.15
Lexical Disambig.
+1.00
+0.85
-1.27
+1.29
+1.78
+1.41
+1.23
+1.59
-1.02
-0.95
Causal Reasoning
+1.00
+0.79
—
+1.38
+1.82
+1.42
+1.38
+1.36
-1.75
-1.60
Pragmatic Inference
+0.17
+0.74
-1.75
—
+1.62
+1.22
+1.13
+1.88
—
—
Theory of Mind
+0.50
+0.65
+1.21
—
+1.06
—
+1.20
+1.48
-2.19
-2.05
Goal Conflict
+1.00
+0.88
-1.33
+1.62
+3.00
+1.61
+1.61
+3.35
-2.65
-2.40
Self-Knowledge
+0.50
+0.91
+1.52
+1.85
+1.66
+1.84
+1.83
+1.45
-1.04
-0.90
Uncertainty Handling
+0.25
+0.55
—
—
—
—
—
—
—
—
Behavioral
Probes
Attn SAE
MLP SAE
Resid SAE
Skip-TC
Skip-TC+Affine
Crosscoder
CLT
CLT+Affine
Feature Relevance (+promotes / −suppresses target)
3
2
1
0
−1
−2
−3